The Matthew Effect in Science Funding

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Abstract

A classic thesis is that scientific achievement exhibits a "Matthew effect": Scientists who have previously been successful are more likely to succeed again, producing increasing distinction. We investigate to what extent the Matthew effect drives the allocation of research funds. To this end we assembled a novel dataset containing all review scores and funding decisions of grant proposals submitted by recent PhDs in a 2 billion Euro granting program. Analyses of review scores reveal that early funding success introduces a growing rift, with winners just above the funding threshold accumulating more than twice as much research funding (€180,000) during the following eight years as non-winners just below it. We find no evidence that winners' improved funding chances in subsequent competitions are due to achievements enabled by the preceding grant, which suggests that early funding itself is an asset for acquiring later funding. Surprisingly, however, the emergent funding gap is partly created by applicants, who after failing to win one grant less often apply for another.

Significance Statement

Why do scientists with similar backgrounds and abilities often end up achieving very different degrees of success? A classic explanation is that academic achievement exhibits a "Matthew effect": Early successes increase future success chances. We analyze novel data from a large academic funding program that present a unique opportunity to quantify the Matthew effect and identify generative mechanisms. Our results show that winners just above the funding threshold accumulate more than twice as much funding during the subsequent eight years as non-winners with near-identical review scores that fall just below the threshold. This effect is partly caused by non-winners ceasing to compete for other funding opportunities, revealing a "participation" mechanism driving the Matthew effect.

Why is academic success so unequally distributed across scientists? The theory of the Matthew effect identifies a self-reinforcing dynamic in academic stratification borne out of the tendency for a scientist's past success to positively affect success in the future (*1*). The theory goes that if only one of two equally talented, young scholars was given an award, the award-winning scholar would go on to have the more successful career. This would happen because the winner would enjoy resource and status advantages over the non-winner (*2-4*). These advantages would lead differences in future success to further grow, setting in motion a cumulative advantage process of increasing distinction (*5-10*). To the extent that luck plays a role in early academic success (*11*), the Matthew effect may undermine meritocracy by allowing an initially fortunate scientist's recognition to self-perpetuate, while an equally talented but initially less fortunate counterpart remains underappreciated (*1,5-7*).

An interdisciplinary literature finds patterns in observational data that are mostly consistent with the Matthew effect (*7*). Empirical testing of the Matthew effect has, however, been hampered by the possibility that observed cumulative differences in achievement are instead the gradual manifestation of unequal academic promise, which can only be imperfectly measured. While an empirical pattern of academic recognition repeatedly going to the same scientists is consistent with the Matthew effect, it is confounded with interpersonal differences in talent or productivity (*12-19*). Namely, increased differentiation in publication, citation, award and employment records may point to endogenously emergent inequality (*3, 20-25*), but can also be interpreted as delayed revelation of variable ability or effort (*5, 7, 10, 13*). Measures of academic success follow extreme distributions (*20-22,26-29*), consistent with the Matthew effect thesis, but these may also reflect a convex correspondence between a scientist's capability and repute, with only the very best receiving the lion's share of recognition (*5, 7, 8, 10, 18, 19, 31-34*).

Our contribution is three-fold: First, we address the causal inference problem using a regression-discontinuity approach. Second, we systematically study the Matthew effect in science funding, an important outcome that has received scant attention. And third, we identify a *participation* mechanism driving the Matthew effect whereby early-stage failure inhibits participation in further competition through discouragement and lack of resources.

We argue that the Matthew effect may be particularly dominant in the accumulation of individual research funding. In contrast to scientists judging the quality of research papers, reviewers of grants, especially personal grants, are often explicitly tasked with evaluating the ability and promise of the applicant, using past achievements as criterion. The *status* mechanism theorized in earlier work, whereby past successes aid quality assessment under uncertainty, should thus be prominent in the allocation of research funds. In addition to status, grants provide a *resource* that can be invested to improve the quality of subsequent work (*1, 24*). Studying science funding also enables us to evaluate the proposed *participation* mechanism, whereby funding success promotes continued participation in future tournaments for funding. This mechanism is particularly powerful in the earliest stages of the scientific career, when young scholars make defining career choices that largely depend on their prospects of acquiring permanent university positions. Winning an early grant will motivate scholars to compete for funding again if they suspect that their chances in future competitions have improved as a result.

Data

We perform a critical test of the Matthew effect in early-career academic funding in the Netherlands. This empirical context provides an ideal research site. A single granting program, the Innovation Research

Incentives Scheme (IRIS), is the primary funding source for young Dutch scientists. Through this program, the Netherlands Organization of Scientific Research (NWO) has since 2000 concentrated €2,400,000,000 in the hands of a minority of 3,660 applicants with proposals deemed of superior promise. We exploit the feature of the program that funding decisions are made using a hard threshold in evaluation scores, with all winners having higher scores than all non-winners. This policy makes a regression discontinuity design possible because differentiation among scores near the threshold can be considered "as good as random" (*36*). We isolate the effects of recent PhDs winning an early-career "Veni" grant by comparing the subsequent funding success of non-winners with evaluation scores just below the threshold to winners with scores just above it. An additional advantage of this research site is that data on applicants in the mid-career "Vidi" competition allow us to assess whether improved rates of success among winners of earlier grants in later contests result from increased participation in these contests. Since the data track many careers for over a decade, we can evaluate the cumulative impact of early-career success.

Results

To identify the effect of winning an early-career grant on winning a mid-career grant we calculate the percentage of mid-career grant winners among early-career grant applicants at different ranks away from the early-career funding threshold (Fig. 1 left panel). Those who won an early-career award by the smallest margin (+1/+2) are roughly two-and-a-half times more likely to win a mid-career award than those who fell just short of winning an early-career award (-1/-2) (exact test, $p < .001$). Parametric estimation (*SI Appendix*) identifies an effect of the same magnitude and significance. Mid-career award chances do not differ between any other pair of neighboring ranks of applicants in the early-career competition, suggesting that this cumulative advantage effect does not spuriously reflect superior proposal quality or scientific ability. The effect is found for both women and men and is robust across academic fields (*SI Appendix*).

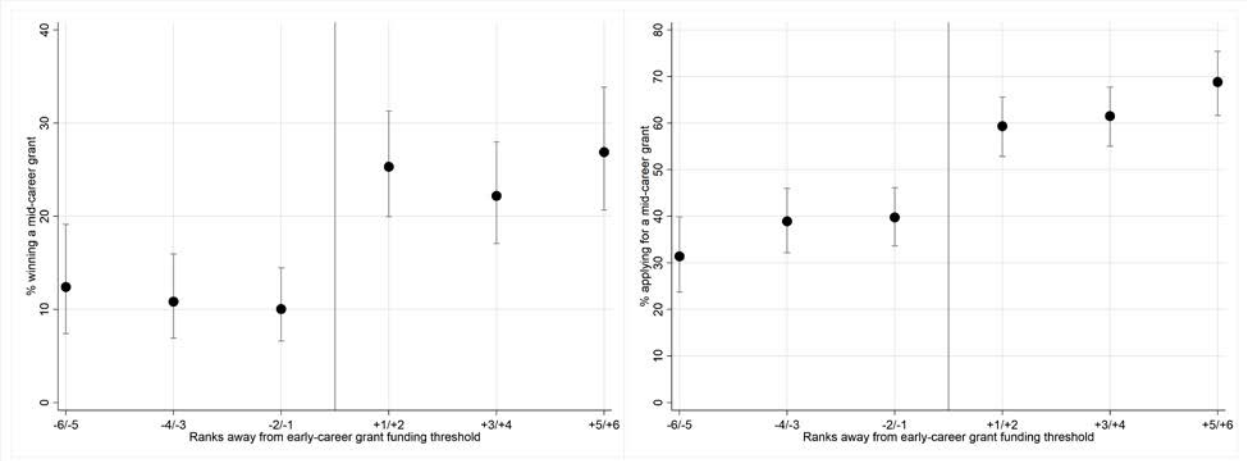


Fig 1. Effect of an early-career grant on winning a mid-career grant (left panel) and on applying for a mid-career grant (right panel). Shown is the percentage of early-career grant applicants winning a mid-career grant (vertical axis in left panel) and applying for a mid-career grant (vertical axis in right panel) for different evaluation ranks of applicants in the early-career grant competition (horizontal axis). Applicants with positive ranks (+) to the right of the funding threshold (vertical line) received an early-career grant, while applicants with negative ranks (-) did not. For each estimate an exact 95% confidence interval is displayed.

To test the presence of the proposed participation mechanism, we calculate the probability that early-career grant applicants of different application ranks submit a proposal to the mid-career competition (Fig. 1, right panel). We find no significant difference between neighboring pairs of ranks in the probability of applying for a mid-career grant except between winners and non-winners closest to the threshold (exact test, $p < .001$). Thus, a substantial portion of the increase in the probability of winning a mid-career grant from 10% to 26% resulting from winning an early-career grant (Fig. 1 left panel) can be accounted for by an increase from 40% to 59% in application chances (Fig. 1 right panel). Supplementary analyses reveal that these differences in application chances are not caused by non-winners exiting academia or moving to another country (*SI Appendix*), but rather by them deciding not to partake in later competitions.

In order to test whether the observed positive effects on winning a mid-career grant (Fig. 1 left panel) can be attributed to increased impact of scholarly work enabled by preceding grants, we compare the number of publications, citations, and H-index of early-career applicants of different ranks (Fig. 2). Consistent with prior research (*37-39*), we find a positive association between ranking in the early-career competition and each of the impact measures both before and after the competition. However, none of the impact measures exhibit a discontinuous increase at the cutoff, and the relationship between competition rank and productive achievement is the same before as after the award competition. This finding indicates that the early-career grant does not raise measurable dimensions of scientific output (see *SI Appendix* for analyses further corroborating this conclusion), and therefore does not suggest that winners of early-career grants in later competitions are better evaluated than non-winners for having achieved greater scientific impact as a result of the earlier grant. This result is consistent with the status mechanism proposed in earlier work (*1, 2425*), whereby the improved success chances would be attributable to recognition for having won the early-career grant. It is also possible that time and resources made available by the early-career grant allow winners to write a higher quality proposal for the mid-career competition.

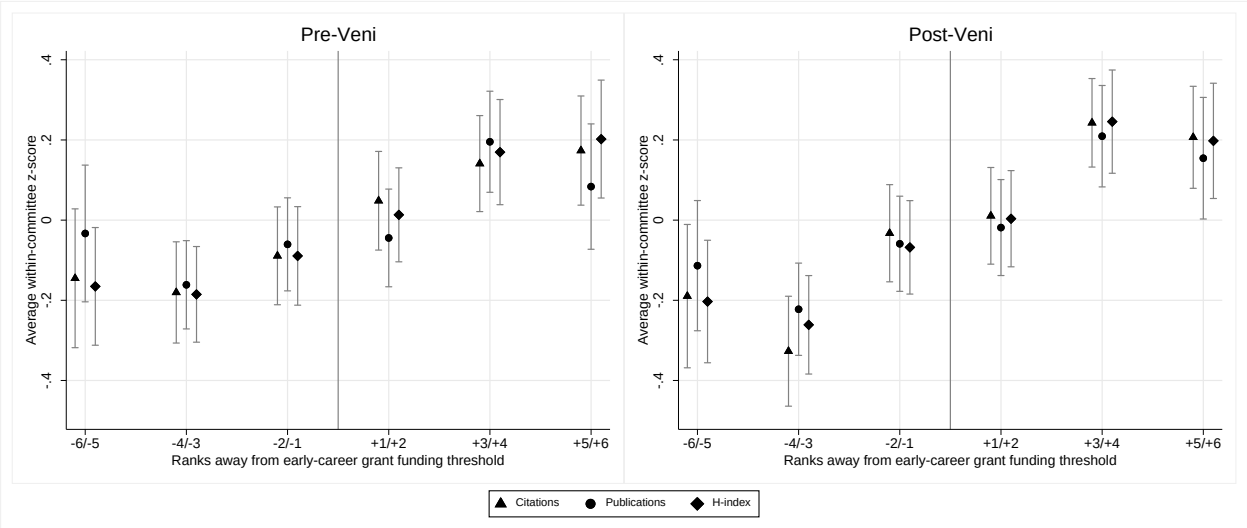


Fig. 2. Scientific impact before (left panel) and after (right panel) early-career grant competition, by ranking. Shown are three measures of scientific impact (vertical axis), the average standardized number of citations (triangles), publications (dots), and H-index (diamonds), for different evaluation ranks of applicants (horizontal axis). Applicants with positive ranks (+) to the right of the funding threshold (vertical line) received an early-career grant, while applicants with negative ranks (-) did not. For each estimate a normal 95% confidence interval is displayed.

To calculate the long-term impact of winning an early-career grant on funding success we pool the scientists' overall success in four major grant competitions: the NWO Vidi grant (max. €800,000), the NWO Vici grant (max. €1,500,000), the ERC Starting grant (max. €1,500,000) and the ERC Consolidator grant (max. €2,000,000). Consistent with the Matthew effect thesis, we find that the head start given to early-career grantees produces a widening gap with non-grantees (Fig. 3). After four years the lowest-ranked winners (+1/+2) have received, on average, €40,000 more in research funding than the best non-winners (-1/-2) (rank-sum test; $p < .01$), and this difference grows to approximately €180,000 after eight years (rank-sum test, $p < .01$). Over the entire period the lowest-ranked winners in the early-career competition accumulated more than twice as much grant money as the highest-ranked non-winners, excluding the €250,000 early-career grant itself. This gap chiefly reflects differences in funding probabilities, as we find the Euro amounts of later awards received by early-career grant winners to equal those of non-winners (*SI Appendix*). In stark contrast, winners with neighboring evaluation ranks do not significantly differ from one another in later funding success, nor do non-winners with neighboring ranks, suggesting that later-stage funding differences between early-career grant winners and non-winners near the funding threshold are not attributable to differences in these scientists' abilities. The emergent funding gap (€180,000) accounts for 40% of the entire difference in accumulated earnings between the best and worst applicant (€450,000) (*SI Appendix*), suggesting that the Matthew effect plays a dominant role in the creation of funding inequality.

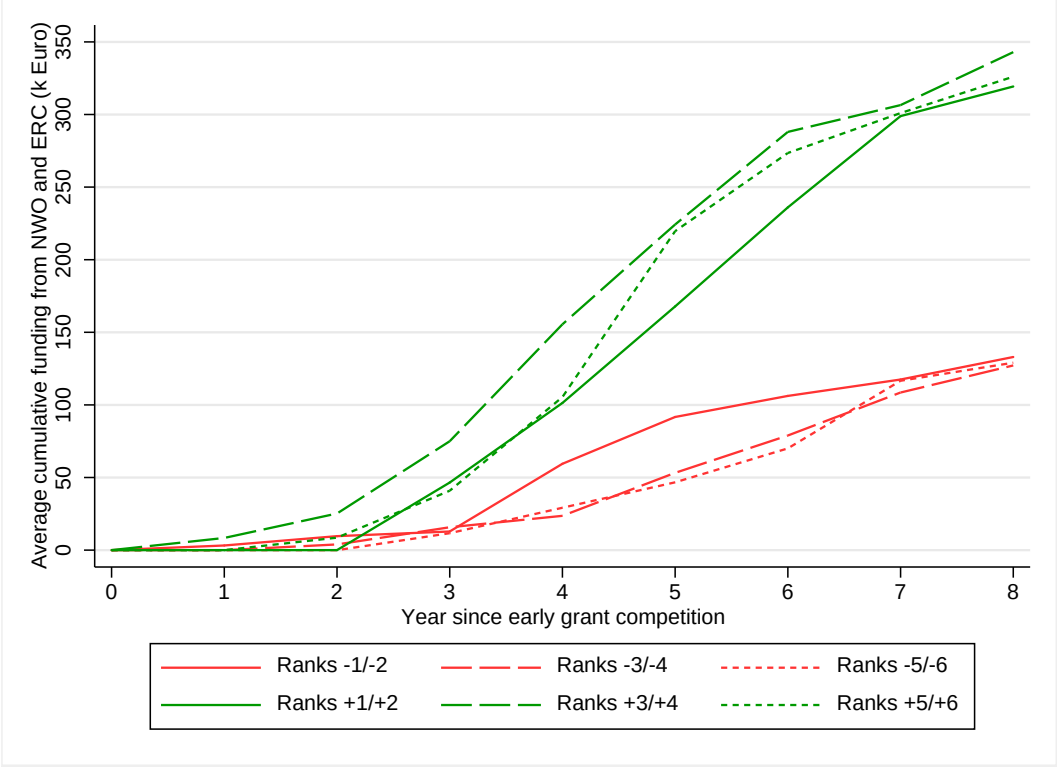


Fig. 3. Accumulation of grant money by early-career grant applicants. Shown is the cumulative amount of funding received in NWO and ERC competitions (vertical axis) as a function of the number of years elapsed since the early-career grant competition (horizontal axis). This relationship is shown for different ranks above (green, +) and below (red, -) the early-career funding threshold.

Lastly, we explore the possibility that early-career grants sort winners into more successful academic career tracks than non-winners. In Fig.4 we find that being awarded an early-career grant raises long-term prospects of a full professorship by 47% (i.e. from 19% to 28% ($p < .05$)). This result confirms that the award competition does not only produce an enduring rift in funding success, but also has a lasting impact on winners' and non-winners' positions in the academic hierarchy.

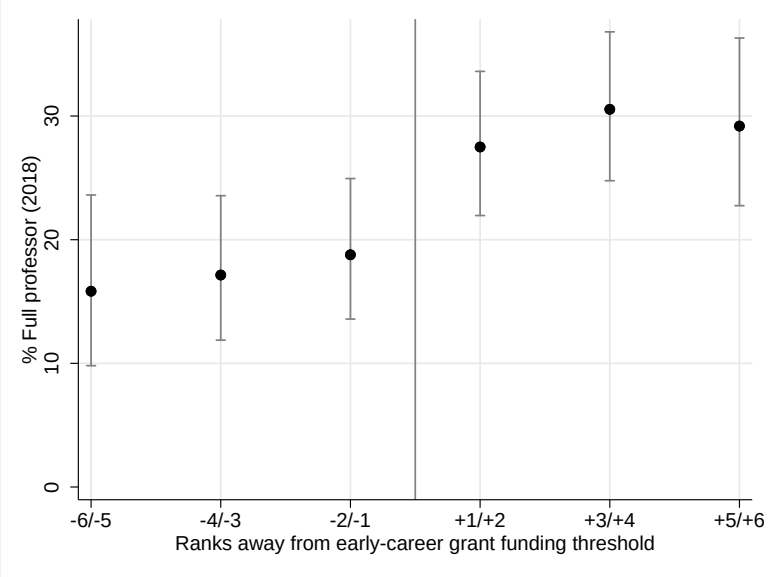


Fig. 4. Effect of an early-career grant on being full professor 10 to 16 years later. Shown is the probability of holding a professorship in 2018 (vertical axis) for different evaluation ranks of applicants in the early-career grant competition (horizontal axis). Applicants with positive ranks (+) to the right of the funding threshold (vertical line) received an early-career grant, while applicants with negative ranks (-) did not. For each estimate an exact 95% confidence interval is displayed.

Discussion

We conclude that funding of early career scientists exhibits a Matthew effect that operates through two mutually reinforcing processes: On the demand side, candidates who won prior awards are evaluated more positively than non-winners, while on the supply side, scientists who were successful in past contests select themselves into applicant pools of subsequent contests at higher rates than unsuccessful scientists.

The key institutional features of the Dutch academic funding system that enable the observed demand side dynamics are widespread in grant competitions elsewhere, which suggest that the positive feedback effect uncovered here is likely to extend to other contexts. Prior academic success is often a merit review criterion, investigators are often unlimited in the number of grants they can pursue, and information on an applicant's past grants is often available for consideration by reviewers, panelists, and program directors. Our results thus raise the question whether funding organizations worldwide should change the common practice of providing information on prior awards and other investigator success metrics to evaluators of applications.

The observed tendency for winners of earlier grants to try their luck in later competitions in greater numbers than non-winners suggests that funding agencies could consider outreach efforts aimed at reducing this gap. One costless measure agencies may take is providing unsuccessful applicants with detailed information on how close evaluation scores were to the funding threshold, which may prevent near-winners with good past proposals from concluding that future odds are too low for investing time and effort in a new application.

Recent studies have documented rising inequality among scientists across the academic world (*39, 40*). Not only do our findings suggest that positive feedback in funding may be a key mechanism through which money is increasingly concentrated in the hands of a few extremely successful scholars, but also that the origins of emergent distinction in scientists' careers may be of an arbitrary nature. This raises the question whether, especially in fields where materials and infrastructure costs are minimal, the distribution of smaller grants across a larger number of scientists could reduce inequality and improve meritocracy without sacrificing efficiency.

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Supporting Information Appendix

Materials and Methods

Table S1 – S9

Fig S1 – S16

References

Supporting Information Appendix

The Matthew Effect in Science Funding

Thijs Bol Mathijs de Vaan Arnout van de Rijt

April 17, 2018

Description of Innovation Research Incentives Scheme

In the main text we analyze data from the Innovation Research Incentives Scheme (IRIS) from the Netherlands Organization of Scientific Research (NWO). The IRIS consists of 3 grants: the Veni-grant (early career, up to 3 years after PhD), the Vidi-grant (mid-career, up to 7 years after PhD), and the Vici-grant (late-career, up to 15 years after PhD). The scheme started in 2000 with the Vidi-grant. In 2002, the Veni- and Vici-grant were added. For each grant there is one funding round per year.[1] An overview of the IRIS-scheme is provided in Table S1.

The main question we aim to answer in this study is whether winning an early career Veni-grant increases one's chances of also winning a mid-career and late-career grant. For the years in which we study Veni-applications (2002-2008), applicants were allowed to apply for the Veni-grant up to 3 times. Veni-grant winners receive €250k, which they can use to fund three years of full time research or four years of 0.75fte. The IRIS-scheme is the main instrument by which NWO allocates public funding to talented scientists in the Netherlands. Moreover, it is the only scheme that does this on an individual basis. There are a few other funding opportunities for scientists in The Netherlands, but the IRIS is the only Dutch funding program that allocates large grants directly to talented individual researchers. Applications to the Veni-program (see Table S3) have risen and the acceptance rate has gone down. In 2015 (the latest available year), 14.3% of the 1,142 applicants were awarded a Veni-grant.

Table S1: Description of IRIS

	Veni	**Vidi**	**Vici**
Start of program	2002	2000	2002
Maximum total amount	€250k	€800k	€1.5m
Maximum years since PhD	3	8	15
Length of grant	3 years	5 years	5 years
Hiring of personnel possible	No	Yes	Yes
Number of awarded grants	±150	±85	±30
Number of applications (2015)	1,124	572	215
Acceptance rate (2015)	14.3%	15.2%	14.9%

Evaluation of Veni applications

The first stage of the Veni-grant application process requires applicants to submit a full proposal. Evaluation of these proposals takes place within eight research divisions (Table S2). This implies that applications to different NWO-divisions are not directly competing with one another for the same pool of funding. NWO-divisions are not of equal

[1] In 2002, 2004, and 2006 there were 2 Veni-rounds.

size, and the number of Veni-grant applications submitted to these divisions varies substantially. The final column in Table S2 shows the mean number of applicants for the Veni grant in the period that we study. Social Sciences and Health Research have many more applications than Physics or Technical Sciences. As a result of variation in the number of applications per division, divisions are organized differently: while smaller divisions have one committee that evaluates applications, larger divisions have multiple committees. The Social Sciences division, for example, has four committees each of which focuses on different disciplines: Economics and Business, Behavior and Education, Law and Politics, and Social Sciences. These four subcommittees work independently from each other and their scores are grouped together after review scores have been assigned. The Physics division, in contrast, has only one committee. Each application is evaluated by about 10 committee members. Committees are appointed by the scientific board of each NWO-division. In the period that we study (2002-2008), committees score the applicants on two dimensions: quality of the researcher (50%), and quality of the proposal (50%). The aggregate of these two factors, averaged over all committee members, constitute the priority score of the applicant.

Table S2: NWO-divisions

NWO-division		**Disciplines (examples)**	**Domain**	μ **applicants**[a]
Earth and Life Sciences	ALW	Biology, Neuroscience	Life	43.9
Health Research	ZonMW	Medicine, Psychiatry	Life	91.9
Chemical sciences	CW	Chemistry, Biochemistry	Beta	19.2
Physical Sciences	EW	Mathematics, Computer Science	Beta	23.9
Physics	N	Physics	Beta	11.5
Technical sciences	STW	Engineering	Beta	20.7
Humanities	GW	Languages, History, Philosophy	Alfa/Gamma	55.1
Social Sciences	MaGW	Sociology, Economics, Psychology	Alfa/Gamma	108.8

[a]The mean number of applicants over the ten Veni-rounds that we study between 2002 and 2008.

Evaluation takes place in three rounds: (a) pre-selection, (b) external reviews, and (c) interviews. First, committees make a pre-selection of applications that are sent out to external reviewers.[2] After the first round of selection, the proposals are sent to external reviewers (2 or 3 for each Veni-proposal) who evaluate the application (ranging from A+ to U [not eligible for funding]). External reviewers are selected by staff members of NWO - not by the committee - and external reviewers are anonymous for both the applicants and the committee. Applicants have one week to write a two-page rebuttal in which they can reflect on issues raised by the external reviewers. The committee then reads the application, reviews, and the rebuttal, and ranks the remaining candidates. Committee members do not blindly follow the scores of the external reviewers, and the correlation between the external reviewer scores and the committee scores is only 0.51 [1].

[2]This is only done when the amount of proposals that is submitted is four times as large as the number of grants that can be awarded.

If there are N available funding slots within a focal NWO-division, the division typically invites up to N*2 applicants for the final stage in the application process: the interview. After the interview stage, committees make a final ranking, giving each candidate a final priority score ranging from 9 (lowest ranking) to 1 (highest ranking). All individual committee members give a score, and the final priority score is the committee average. Committee members then discuss the final ranking. These discussions typically focus on applications in the "gray area", and sometimes committees decide to rerank applicants. Qualitative studies have found that in the IRIS, committee members find it almost impossible to differentiate between applicants that are just above or below the cutoff [2, 3].

Once the rankings of the NWO-divisions are finalized, the NWO-board compiles a list of winners. NWO does not communicate to applicants what their final ranking is. This means that applicants do not know if they ended just above or below the threshold - all they know is whether they won or lost.

Data and data quality

Data from IRIS

Our main dataset consists of all application information from the IRIS. The dataset is constructed from two sets of files. First, a database file that contains demographic information for applicants per application (name, title, sex, institution, birth date). Second, a series of ranking-files that contain the final ranking of the candidates for a grant, round, and NWO-division. These files contain information on the applicants (name, title) and on the application (final ranking, priority score, grant awarded). The two files are matched by the unique file numbers in the ranking-files and the database-files. This matching between the database-files and the ranking files for the years 2000-2008 was already done by researchers for another project [4]. We use their matched data for these years, but append it with information for priority scores that were missing for the Veni rounds in 2004.

For 1,795 applications (about 9% of all applications), the file number in the database did not correspond with the file number in the ranking-files. These applications are matched manually on first and last name. The quality of the matching is very high, since we match on first name and last name by grant, round, and division. After matching, we end up with a dataset that contains 20,376 applications. The distribution of applications across grants and years can be found in Table S3. About 1% of the applications (221 applications) are dropped because they are duplicates within a grant and round. The duplicates are applicants that switch NWO-division during the application and as a result get a new file number. We only keep the final applications in the correct NWO-division and end up with 20,155 applications for all grants from 2000-2016.

Table S3: Applications to IRIS-scheme

	Veni	**Vidi**	**Vici**
2000	-	153	-
2001	-	155	-
2002	513	431	199
2003	428	275	128
2004	695	271	136
2005	511	309	148
2006	801	349	167
2007	533	433	221
2008	644	384	165
2009	809	503	220
2010	981	557	215
2011	958	652	234
2012	939	431	212
2013	1,002	542	205
2014	1,088	520	217
2015	1,126	575	218
2016	-	-	236
Total	10,915	6,540	2,921

Note: The cell counts refer to the raw number of applications in our data. Duplicate applications (see text) are still included. In 2002, 2004, and 2006 there were 2 Veni rounds. The Veni and Vidi only started in 2002, the Vidi started in 2000 (under a different name). The Veni 2016 and Vidi 2016 rounds are not finished yet, and are therefore not included in our dataset.

For some rounds and grants the rank-files do not include an indicator of who won the grant. To make sure that we have all the correct winners in our data, we use the public announcements on the official website of NWO.[3] We match the winners by grant, round, NWO division, and last name. Whenever a match is not possible (because there are duplicates on these variables), matches are made manually.

The final challenge for the dataset is to identify unique applicants: the 20,155 applications were not submitted by 20,155 different applicants. We identify unique applicants in two steps. First, all applicants with the same last name, sex, and birth date are treated as the same person. Out of the 20,376 cases, we could match 17,031 by using this information. The remaining 3,345 cases were unmatched primarily because of missing information on birth date or because of inconsistent spelling of the last name between applications. These cases were all matched manually by using more information, such as first name, initials, NWO division. For 26 applications (0.1%) we were not sure if the manual match was successful so we dropped them from our analyses. Dropping those cases from the analyses does not affect our results. The final dataset consists of 12,555 unique applicants.

[3]All winners of all grants can be found at `http://www.nwo.nl/onderzoek-en-resultaten/programmas/vernieuwingsimpuls/toekenningen`, last visited April 17, 2018.

Data from ERC program

While NWO is the single provider of large individual grants in the Netherlands, the European Research Council also awards three individual grants. These grants follow a similar pattern as the IRIS, but are larger of magnitude: ERC Starting Grant (€1.5m), ERC Consolidator Grant (€ max 2.0m), and the ERC Advanced Grant[4] (€2.5m). The ERC grant program started in 2007, with the ERC Consolidator Grant only starting in 2013. We match the publicly available data on the winners of the ERC grants[5] from 2007-2016 that reside in the Netherlands to the applicants in the IRIS dataset. We find 375 ERC-winners in our dataset.

Data from Scopus

We match publication information from Elsevier's Scopus to our dataset. Here we only match the information to our analytical sample (see below) of 1,428 applicants that progressed to the interview in the Veni-grant rounds from 2002 and 2008. We created a program that interacts with the Scopus API and returns a list of potential Scopus IDs for a combination of last names and first names from our data. When the first name is missing, we use initials. For about 50% of the cases only a single match is found. For the rest of the cases, multiple Scopus IDs are returned. Often this is caused by single authors having multiple IDs in Scopus. For the other cases, we use information on sex, research topic (derived from the name applications), and home institution to make manual matches. For about 145 applicants, the program did not find any match. For these cases, we added the Scopus ID manually. In total we were able to match a Scopus ID for 1,390 applicants. For 38 cases we were unable to find a Scopus ID. An extensive web search indicated that 25 of them did not publish anything in a journal that is indexed in Scopus, or that they left academia. We were unable to find reliable information on 13 applicants (0.9%). They are removed from the analyses that use Scopus data.

Professor data

Finally, we match information on the academic position of the applicant to our data. For all applicants around the cutoff (-6 to +6) we performed a web-search. In most cases, professorships are publicly announced on university websites and distinguishable from assistant and associate professorships by the title "prof." In the Netherlands this title can only be used by full professors. In our analyses both regular professors and professors by special appointment are coded as 1. Using information on university websites we have established which applicants are full professor by the time of data-gathering (January 2018). For 5 from the 1,255 applicants we found ambiguous information on their current position. They are excluded from these analyses.

[4]No one in our sample (Veni applicants 2002-2008) has won an ERC Advanced Grant.

[5]All the winners of the ERC grants can be found at `https://erc.europa.eu/projects-figures/erc-funded-projects`, last visited April 17, 2018.

An overview of all the main variables that we use in the analyses can be found in Table S4.

Table S4: Overview of variables

Variable	Description	Source
Rank	The rank of the applicant in a competition (combination of round and NWO-division). The rank is based on the priority scores (1-9).	IRIS data
NWO-division	The division the Veni-applicant competed in (see Table S2).	IRIS data
Veni round	The round of the Veni-application (2002-2008).	IRIS data
Veni winner	Variable indicating if applicant won the Veni grant.	IRIS data; public announcements
Vidi winner	Variable indicating if applicant won the Vidi grant.	IRIS data; public announcements
Vici winner	Variable indicating if applicant won the Vici grant.	IRIS data; public announcements
Vidi applicant	Variable indicating if applicant applied for Vidi grant.	IRIS data
Professor	Variable indicating if applicant is full professor in 2018.	University websites
Sex	Sex of the applicant	IRIS data
ERC winner	Winner of ERC grant.	ERC public winner list
ERC year	Year of winning ERC grant.	ERC public winner list
Publications	Standardized (z-score) of total publications within NWO-division and round. Pre-Veni publications are all publications from -10 years to the application year. Post-Veni publications are all publications from Veni-application to +5 years.	Scopus
Citations	Standardized (z-score) of ln 1 + total citations within NWO-division and round. Pre-Veni citations are all citations from -10 years to the application year. Post-Veni citations are all citations from Veni-application to +5 years.	Scopus
H-index	Standardized (z-score) of H-index within NWO-division and round. Pre-Veni H-index is the H-index in the year of application. Post-Veni H-index is the H-index at +5 years from Veni application.	Scopus

Sample selection

In our analyses we follow Veni-grant applicants from 2002-2008 to see how likely they were to win the mid-career Vidi-grant and how well they published before and after the Veni-grant application. When the late-career Vici-grant is the outcome (see below), we only use applicants from 2002-2006, because Veni-grant winners from 2007 and 2008 are unlikely to apply for a Vici grant before 2016.

There are two main reasons to restrict our sample of Veni-grant applicants to applicants that have applied prior to 2009. First, later applicants have fewer opportunities to apply for a Vidi-grant. The Vidi-grant can be applied for up to 8 years after the PhD (see Table S1). This means that someone who graduates in 2008 (and potentially applies for a Veni-grant in 2009) is eligible to apply for a Vidi-grant until 2016 (the most recent Vidi cycle in our data is 2015). Using this strategy, we are able to observe all potential

applications to the mid-career Vidi-grant for the Veni-cohorts within our sample: the last year of Vidi-grant eligibility for the last cohort of Veni-grant applicants was 2015.

Second, after 2008, NWO implemented some changes to the IRIS. Until 2008, universities and research institutes had to match the grant that an applicant received with their own funds. NWO paid two thirds, whereas the other third was paid by the employer. The matching was paid from research funds that the Ministry of Education, Culture and Knowledge allocated directly to universities ("first money stream"). From 2008 onwards, about €100m was moved from direct government funding to indirect government funding ("second money stream"), thereby increasing the budget of the IRIS [5]. Moreover, from 2009-2011 NWO used a different evaluation protocol for the Veni-program. In those years, each NWO-division awarded two thirds of the grants that they were allowed to award directly, but the other third was awarded through a broader competition across the research divisions.

From 2002 to 2008 there have been 10 Veni-rounds. We analyze all 8 NWO-divisions from which we use their rankings. We end up using 77 rounds because for the Physics-division the number of non-winning interviewees was too low (0) for both rounds in 2002 and the first round in 2004.

In our analyses we only focus on applicants that advanced to the interview round. The main reason for this is that our identification strategy uses the discontinuity between the lowest scoring winners and the highest scoring non-winners. It therefore only makes sense to look at Veni-applicants who advanced to the interview stage. The second reason is that within each division, a new ranking is being made for the applicants that advance to the interview stage. In each round, a new ranking is being made. This means that we cannot compare the scores of the highest scoring applicants that did not advance to the interview round to the lowest scoring applicants after the interview. Table S5 describes the number of applications around the cutoff that we analyze.

Table S5: Cases in our analytical sample

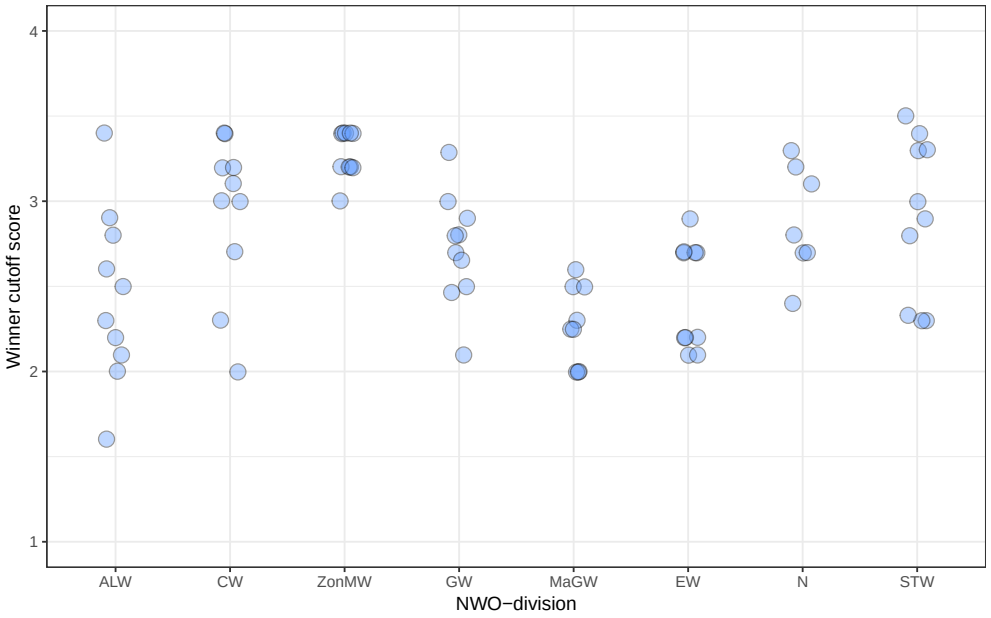
Veni (2002-2008)	
Rank	# of Applicants
-6/-5	137
-4/-3	203
-2/-1	249
+1/+2	241
+3/+4	239
+5/+6	186
Total	1,255

Note: The cell counts refer to the number of applications that we analyze in each design.

Methods

We use non-parametric methods and compare the highest-scoring non-winners to the lowest-scoring winners. In particular, we use the rankings of the lowest ranked winners and the rankings of the highest ranked non-winners as the forcing variable in a Regression Discontinuity (RD) design. The main reason to use the final ranking instead of the priority score is because there are substantial differences between NWO-divisions and between rounds within NWO-divisions in the priority scores that are assigned (see Figure S1). However, the priority scores are the sole determinants of the final ranking. Tied priority scores are assigned the same rank, so there may be multiple applicants at the same rank. If there are, for example, multiple highest scoring non-winners, these highest scoring non-winners are all ranked at -1. In our analyses we only include those committees for which we have at least one winner (+1) and one non-winner (-1). This is the case for all competitions, except for three early competitions in the smallest NWO-division, Physics (see above).

Figure S1: Winner cutoffs by NWO-division and round



Note: Each point refers to the cutoff on the priority score used to differentiate winners from losers for different rounds and NWO-divisions (Veni 2002-2008).

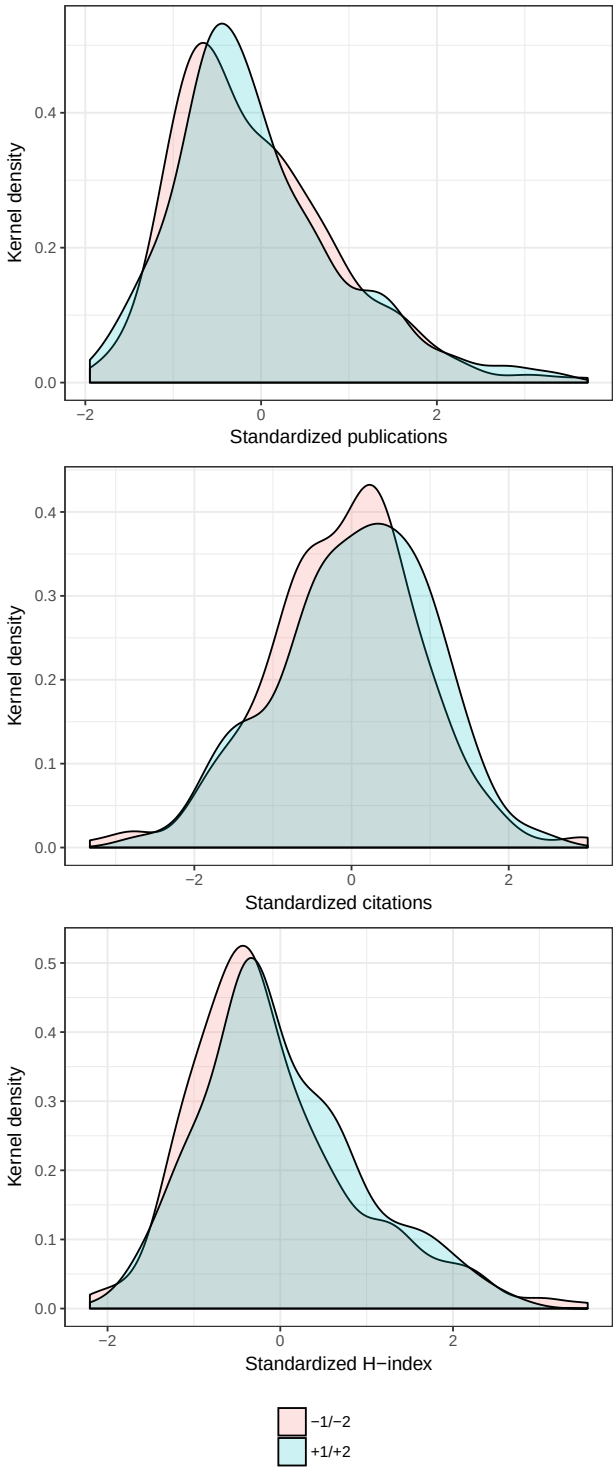
In Figure S1 we plot the score of the lowest-scoring application that was awarded a grant for every NWO-division. The highest possible score is 1 while the lowest possible score is 9. A point at 2.6 does therefore imply that all winners must have had a final priority score between 2.6 and 1. Figure S1 shows substantial variation in the priority score that separates winners from non-winners. Earlier research shows that this is because committee members are not focused on the absolute priority scores of the applicants, but

more with their relative ranking. An important reason to use the final ranking instead of the priority scores is that committee members discuss "whether candidate A rightfully [stands] above candidate B, disregarding the better score of A" [2, p. 122-123].

To identify the causal effect of winning an early-career grant on winning a mid- and later-career grant, we use the early-career grant ranking in an RD design [6]. The goal of the RD design is to show whether the outcome of interest (i.e. likelihood of winning a mid-career or late-career grant) exhibits a discontinuity at the cutoff in the forcing variable (i.e. ranking of applicants) [7]. For the results of an RD design to be valid, it is important to verify that 1) there are no other pre-treatment variables that exhibit a discontinuity around the cutoff and 2) that there is no manipulation of the forcing variable that invalidates inferences.

The first assumption we should verify is whether there are no large differences between applicants that just won and applicants that just lost a Veni-grant. Obviously we are only able to check this assumption using observable information. In the main text of this article, Panel A of Figure 3 has already shown that there is no discontinuity in the number of publications, the number of citations, and the H-index around the cutoff. Here we explore those results in more detail. Since there are substantial differences between disciplines in publication and citation cultures, we calculate the z-score for each applicant within his or her NWO-division and round. This implies that each applicant's score captures his or her relative position with respect to peers in that same round in the same committee. The z-score is calculated over applicants that fall within the range that we observe in the main text (-6 until +6). Since the distribution of citations is heavily skewed, we use the natural logarithm of citations instead of the raw count. We add 1 citation to the total citation count in order to be able to perform a log-conversion for applicants with zero citations [6].

Figure S2: Kernel density of Pre-Veni academic achievement



In Figure S2 we plot the kernel density plots for the highest ranked non-winners (-1/-2) and the lowest ranked winners (+1/+2). We see that especially for publications and H-index the distributions of these two groups are very similar, confirming the idea that around the cutoff it is virtually impossible to separate between applicants based on their earlier performance. In the distribution for citations we see that winners perform slightly

Table S6: T-tests of pre-Veni indicators

	$\mu_{+1/+2}$	$\mu_{-1/-2}$	Δ	t-statistic	p-value
Publications (standardized)	-0.044	-0.060	0.016	0.187	0.852
Citations (standardized)	0.048	-0.089	0.137	1.552	0.121
H-index (standardized)	0.013	-0.089	0.103	1.181	0.238
Observations	486				

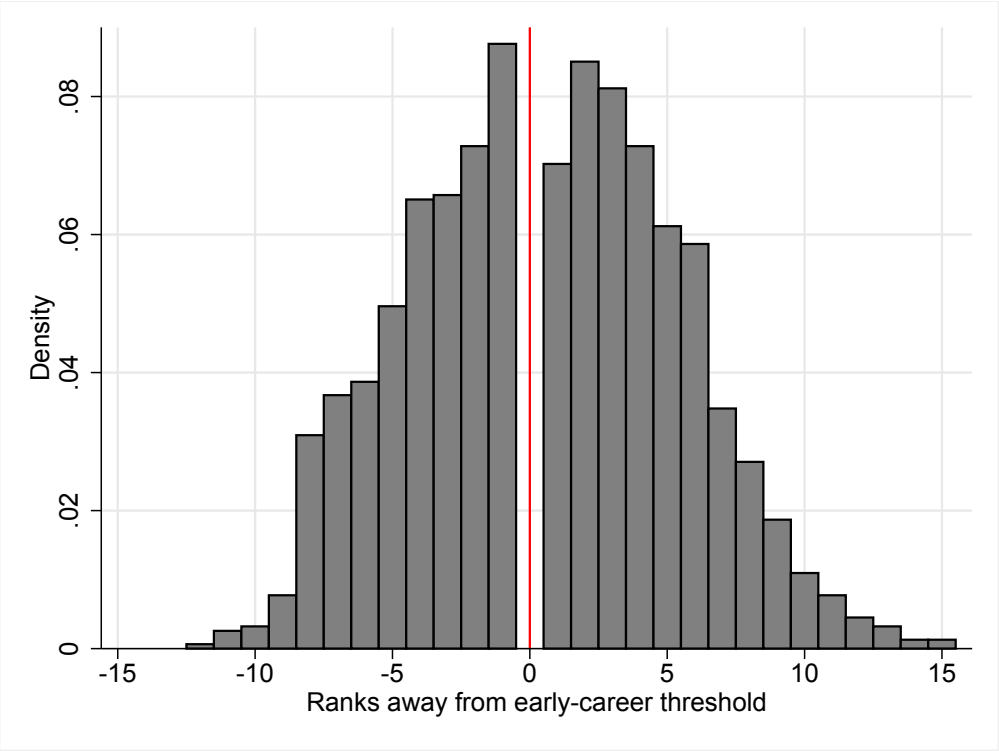
Note: Two-sided t-tests on the difference (Δ) between the means of the lowest ranked winners ($\mu_{+1/+2}$) and the highest ranked non-winners ($\mu_{-1/-2}$).

better, but t-tests confirm that the mean differences between the two groups for all three indicators are not significant in one- or two-tailed tests ($\alpha = 0.05$). The t-tests for the comparisons between the lowest ranked winners and highest ranked losers can be found in Table S6. Jointly, Figure 3 in the main text, Figure S2, and Table S6 demonstrate that the observed discontinuity in our main variable of interest is unlikely to be caused by a factor other than winning a Veni-grant.

A second test to assess the validity of the results produced by an RD design entails verifying continuity in the density of the covariate that underlies treatment assignment at the cutoff (i.e. the forcing variable), against the alternative of a jump in the density function at that point [7]. The intuition for using this test is that if some individuals are able to precisely position themselves just above the cutoff, these individuals are not readily comparable to those just below the cutoff. As a result, any observed discontinuity in the outcome variable may not result from treatment, but rather from some (un)observed difference between cases on either side of the cutoff.

In the setting of science funding in the Netherlands, there is qualitative evidence that suggests that the position of individuals around the cutoff is affected by some manipulation. In particular, there are often multiple applicants with ex aequo priority scores around the cutoff, but since there are fewer funding slots left, only a few of these tied applicants can get funded. Since NWO strives for gender equity this selection process is often not random but based on gender, i.e. with equal rankings, women are prioritized over men. For example, if there are N applicants around the cutoff and if there are only N-X grants left to award, women in the group of N applicants will be prioritized over men.

Figure S3: Density of applicants

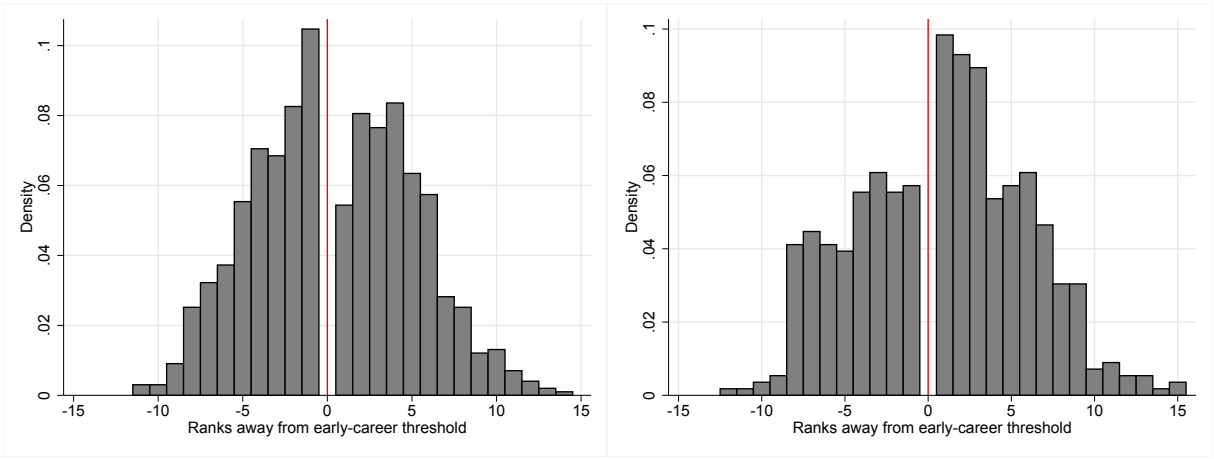


Density of candidates over ranks for all Veni-rounds 2002-2008.

Figure S3 shows a density plot of the forcing variable which indicates that there is a small drop in the number of cases right after the cutoff. Although the discreteness of the forcing variable prevents a McCrary density test [8], the different fractions for rank -1 and rank 1 in Figure S3 is suggestive of manipulation around the cutoff. The pattern is consistent with the qualitative process described above: from a group of applicants with ex aequo priority scores, a small number of applicants is selected to receive the few remaining grants. To inspect this issue in more detail, Figure S4 shows similar graphs but broken up by gender. Figure S4a shows the density plot for male applicants and the difference between the fractions at rank -1 and rank 1 is more pronounced than in Figure S3. Figure S4b shows the opposite pattern: the difference between the fractions at rank -1 and rank 1 is in the opposite direction. The patterns shown in these figures are consistent with funding slots being filled with women who had scores equal to men that were not awarded the grant.

Despite this discontinuity in the density of the forcing variable around the cutoff, the use of an RD design to test the effect of winning a grant (treatment) on applying for future grants and winning future grants remains valid. If the initial rank is correct (i.e. better quality applications are ranked higher), but gender quotas cause applications of similar quality to receive different ranks, similar quality applications will end up on both sides of the funding cutoff. In other words, shifting applicants based on gender around the cutoff

Figure S4: Density of applicants by gender



(a) Density of male applicants

(b) Density of female applicants

is unlikely to cause inflated effect sizes. Moreover, additional analyses (see Figure S7) indicate that the main results are similar for men and women.

Robustness of results to specific design choices

For the analyses presented in the main text of the paper we had to make several choices regarding our research design. In this section we explore whether our findings are robust against alternative research design choices.

Parametric RD analysis

In Figure 1 we present a graphical overview of our RD-estimate, only focusing on the observations very close to the cutoff. Since there are zero winners who received committee scores higher than a non-winner, a sharp RD design is warranted. Table S7 shows the RD-estimate using different bandwidths and specifications. We mainly focus on -4/+4 as this is the optimal bandwidth as estimated with the Imbens and Kalyanaraman bandwidth estimator[9]. The models are estimated Linear Probability Models using OLS for ease of interpretation but the results are identical when estimated using logistic regression. The logistic regression estimates are shown in panel B.

The first effect is obtained from a model where we only include a dummy-variable for winning, without using our forcing variable (the rank away from the cutoff). This coefficient (0.134) therefore is a simple comparison of winners to non-winners in the range -4 to 4. Within that bandwidth Veni-winners are 13.4 percentage points (*pp*) more likely to win a Vidi. In the second model we add a linear term for *rank*. The intuition here is that an average difference between winners and non-winners might be a consequence of a linear trend in rank (-4 are very unlikely to win, +4 very likely to win). However, including a linear term for rank does not affect our estimate of the effect of winning a Veni

on winning a Vidi (0.147 or 14.7 *pp*). The next four models include a linear and quadratic term for our forcing variable to allow the rank variable to have a different functional form. We gradually increase the bandwidth from -3/+3 to -6/+6, but our RD-estimate remains statistically significant and is fairly stable in size. The last model in the table takes the model with our most conservative estimate of the effect (using the -6/+6 bandwidth and *rank* and $rank^2$) and includes division fixed effects for the divisions listed in Table S2.

Table S7: RD regression with different bandwidths

Panel A: Linear Probability Estimates

	w/o *rank*	with *rank*	with *rank* and $rank^2$				
	-4/+4	-4/+4	-3/+3	-4/+4	-5/+5	-6/+6	-6/+6
RD-estimate	0.134**	0.147*	0.184*	0.147*	0.141**	0.132**	0.125**
	(5.52)	(2.49)	(2.52)	(2.46)	(2.73)	(2.87)	(2.66)
Observations	932	932	718	932	1104	1255	1255
Division FE	No	No	No	No	No	No	Yes

Panel B: Logistic Regression Estimates

	w/o *rank*	with *rank*	with *rank* and $rank^2$				
	-4/+4	-4/+4	-3/+3	-4/+4	-5/+5	-6/+6	-6/+6
RD-estimate	0.987***	1.089*	1.243*	1.136*	1.122**	1.051**	1.015**
	(5.26)	(2.49)	(2.22)	(2.50)	(2.84)	(2.97)	(2.84)
Rank		-0.021	-3.767	-2.467	-2.699	-1.171	-0.247
		(-0.26)	(-0.49)	(-0.36)	(-0.42)	(-0.19)	(-0.04)
$Rank^2$			-2.540	1.097	2.555	2.965	2.833
			(-0.87)	(0.39)	(0.92)	(1.07)	(1.01)
Intercept	-2.147***	-2.203***	-2.335***	-2.231***	-2.195***	-2.141***	2.201***
	(-13.98)	(-8.97)	(2.52)	(-8.41)	(-9.31)	(-9.92)	(-7.60)
Observations	932	932	718	932	1104	1255	1255
Division FE	No	No	No	No	No	No	Yes

t statistics in parentheses. The logistic regression coefficients are log-odds.

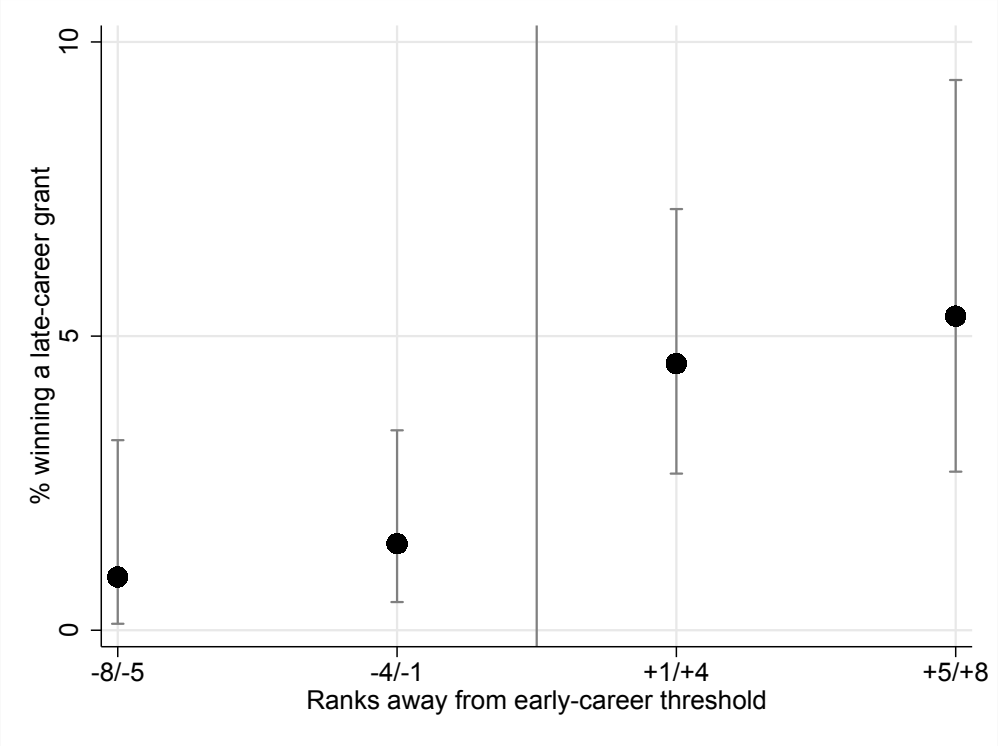
* $p < 0.05$, ** $p < 0.01$

Winning a Vici

Additional analyses indicate that there is an effect of winning a Veni-grant on winning a Vici-grant. Since the Vici-grant is a late-career grant, we can only analyze the Veni-cohorts of 2002-2006. Applicants in later years are unlikely to have even applied for the

Vici (maximum of 15 years after PhD). Because of the fewer years, we group applicants from four instead of two ranks. Although the baseline probabilities for winning a Vici are lower, we find that Veni-applicants at ranks +1/+4 have about a 3 times larger probability of winning a Vici grant than Veni-applicants at ranks -4/-1. This difference around the cutoff is significant ($p<0.05$), whereas we find no significant difference between other neighboring ranks.

Figure S5: Probability to win a Vici-grant

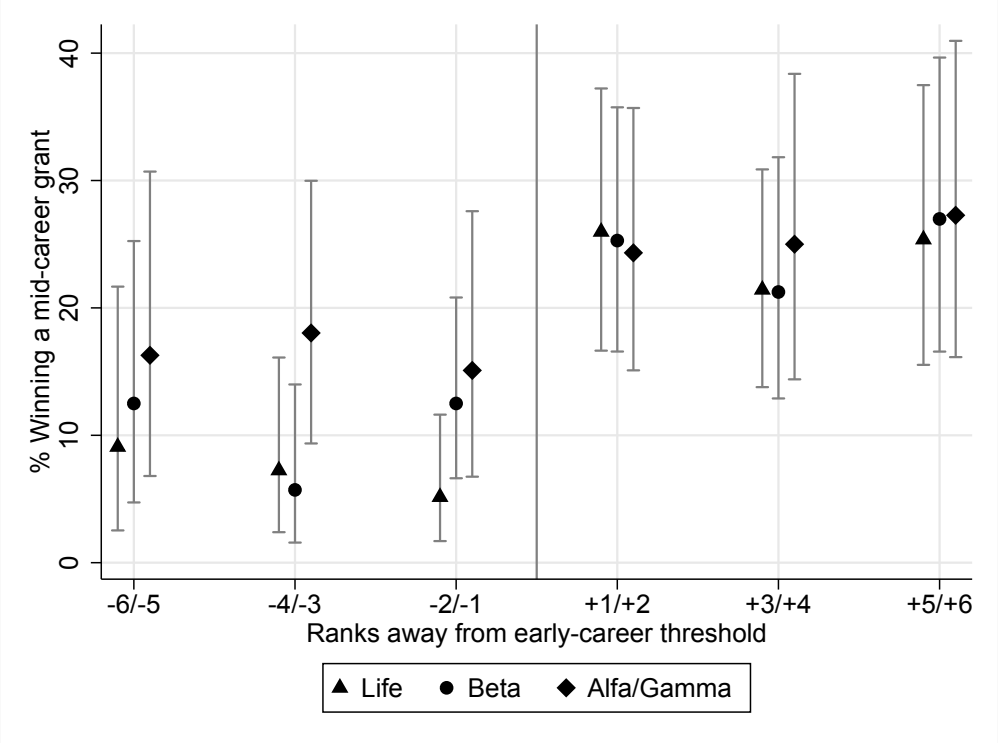


Split up by scientific area and gender

Figure 1 of the paper shows the RD results for data of pooled applications. To show that our findings are not only driven by cases from a single funding division, we break up the results by broad domain in Figure S6. The graph clearly shows that the effect is present in all three divisions. We find no statistically significant differences across domains in the gap between -1/-2 and +1/+2. This suggests that the process we set out to identify (i.e. the Matthew effect) is generalizable across scientific domains.

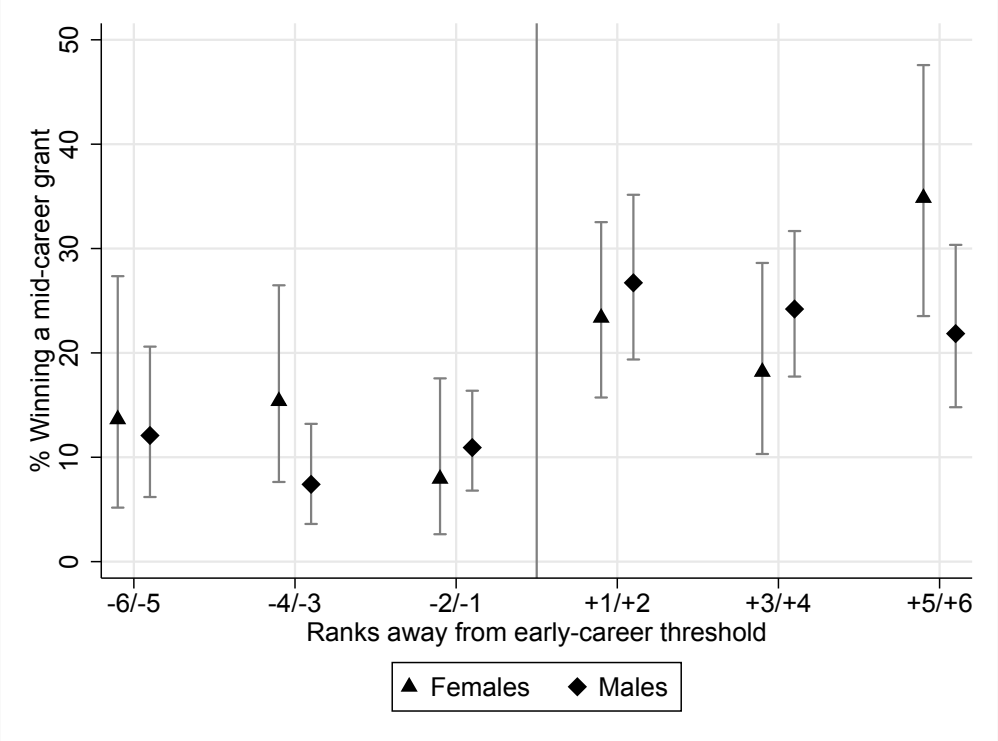
A second obvious dimension that can be used to split up the data is gender. To show that the our findings are not only driven by men or by women, we break up the results by gender in Figure S7. Again, the graph clearly shows that the effect is fairly similar for both men and women.

Figure S6: Probability to win a Vidi (by research area)



Note: See Table S2 for an overview of the clustering of divisions. The sample sizes do not allow a further disaggregation of the data.

Figure S7: Probability to win a Vidi (by sex)



Mediation by research output

In addition to exploring the robustness of our findings by categories (i.e. division and gender), we performed regression analyses to formally test whether the causal effect of winning a Veni-grant on the likelihood of winning a Vidi-grant was mediated by post-Veni differences in academic performance. Table S8 contains the coefficient estimates for a linear probability model that regresses the likelihood of winning a Vidi-grant on winning a Veni-grant. In particular, Model 1 shows that the chances of winning a Vidi-grant increase by almost 16% for winners with the lowest rank compared to non-winners with the highest rank. In Model 2 to Model 5, we stepwise include three additional predictors: the number of Post-Veni publications, the number of Post-Veni citations, and the Post-Veni H-index. Model 5 shows that when we control for these variables neither the effect size, nor the significance level of the variable of interest is affected. And while all three control variables are positively associated with the likelihood of winning a Vidi-grant, only the number of Post-Veni publications is a significant predictor in the final model. When we estimate the same equations using logistic regression instead of linear probability models (Table S9, we find exaclty the same results.

Table S8: Mediation of output variable effects on winning a Vidi (OLS)

	(1)	(2)	(3)	(4)	(5)
+1/+2 (vs. -1/-2)	0.157**	0.153**	0.154**	0.151**	0.151**
	(4.61)	(4.61)	(4.60)	(4.55)	(4.57)
Post-Veni publications		0.083**			0.057*
		(4.21)			(2.43)
Post-Veni citations			0.063**		-0.011
			(3.79)		(-0.39)
Post-Veni H-index				0.078**	0.051
				(4.36)	(1.57)
Constant	0.098**	0.102**	0.100**	0.103**	0.104**
	(5.15)	(5.39)	(5.21)	(5.34)	(5.42)
Observations	486	486	486	486	486
R^2	0.04	0.09	0.07	0.08	0.09

t statistics in parentheses

* $p < 0.05$, ** $p < 0.01$

Accumulation of research funding

In Figure 2 in the main text we show that the average gap in funding money between the lowest scoring winners and the highest-scoring non-winners widens over time. Figure S8 shows the same pattern with confidence intervals. After 8 years the lowest scoring winners have accumulated on average 319k Euro (excluding the Veni), whereas the highest scoring

Table S9: Mediation of output variable effects on winning a Vidi (Logit)

	(1)	(2)	(3)	(4)	(5)
wonvidi					
+1/+2 (vs. -1/-2)	1.148**	1.183**	1.161**	1.153**	1.177**
	(4.40)	(4.42)	(4.39)	(4.33)	(4.38)
Post-Veni publications		0.537**			0.349*
		(4.50)			(2.27)
Post-Veni citations			0.523**		-0.008
			(3.58)		(-0.03)
Post-Veni H-index				0.557**	0.333
				(4.24)	(1.24)
Constant	-2.225**	-2.306**	-2.302**	-2.296**	-2.322**
	(-10.35)	(-10.33)	(-10.39)	(-10.35)	(-10.31)
Observations	486	486	486	486	486

t statistics in parentheses

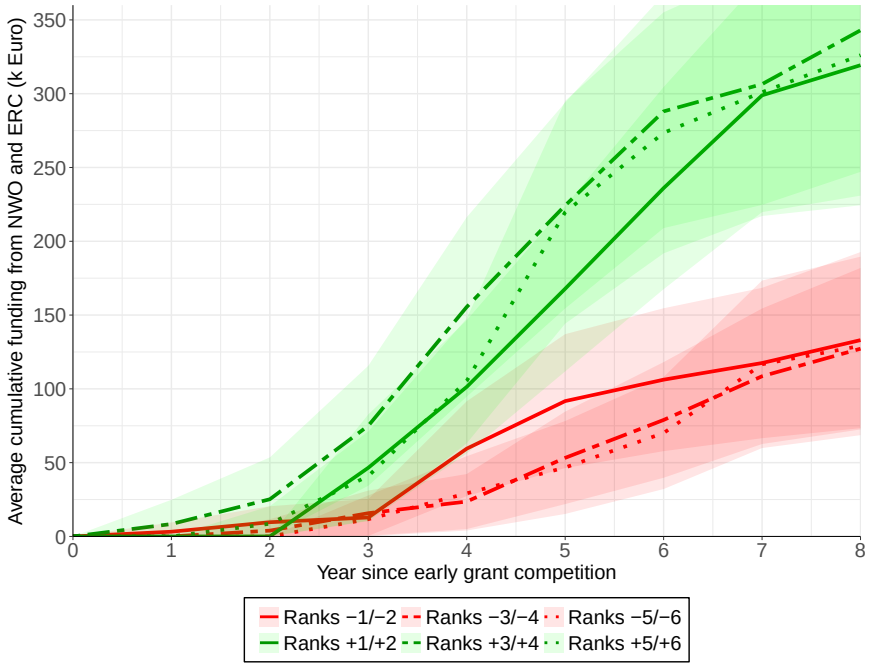
* $p < 0.05$, ** $p < 0.01$

non-winners have obtained 133k Euro. One mechanism that could produce this gap is Veni winners subsequently winning larger grants relative to Veni non-winners. This could happen, for example, if the former propose more applied research than the latter, which requires more funding. To test this we looked at the average size of accumulated ERC grants by the 6 lowest ranked winners and 6 highest ranked non-winners. When winning a grant, the average funding is comparable across the two groups (1564k Euro for non-winners and 1596k Euro for winners, $p = 0.762$ (two-sided t-test)). This shows that the increasing gap between those just above and just below the cutoff is not explained by differences in the size of later grants but rather by their frequency.

Figure S9 also shows the accumulation of research funding, but now only for those just around the cutoff (-1/2 and +1/+2) as well as the highest scoring applicants (the winners) and the lowest scoring applicant[6]. We find that about 40% of the total gap in accumulated research funding after 8 years between the highest and lowest scoring applicants (450k euro) lies between the applicants just above and below the cutoff (180k euro). This highlights the relative importance of the early grant decision for the overall inequality in acquired research funding after 8 years.

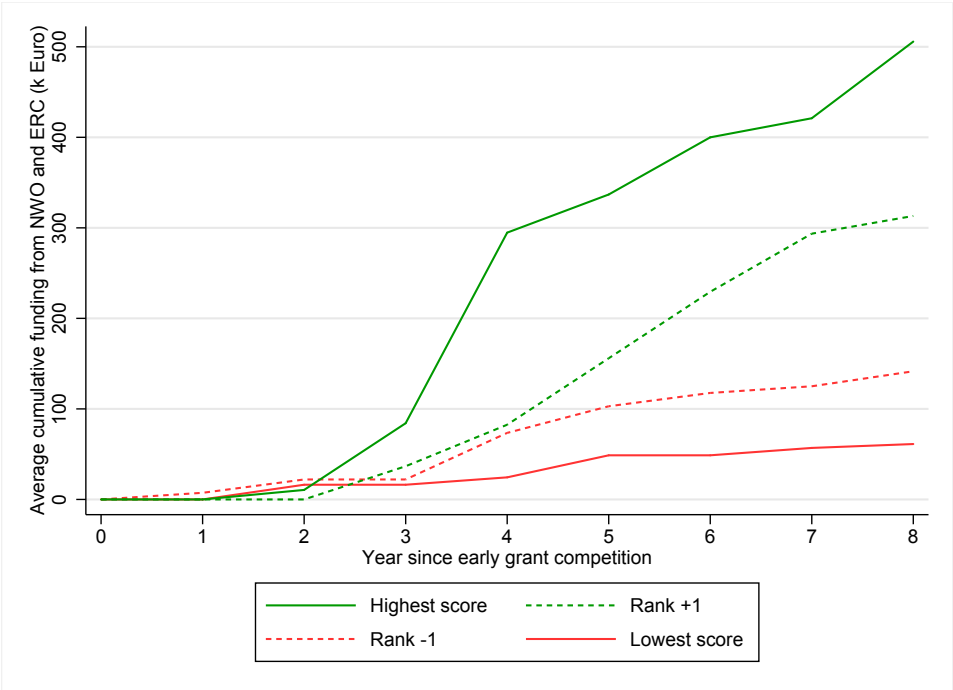
[6]For practically all rounds from 2002-2008 we only have the final scores of applicants that went to the interview, which means that the lowest scoring applicant is almost always the lowest scoring interviewee.

Figure S8: Accumulation of grant money by early-career grant applicants



Note: Shaded areas represent the 95% confidence intervals.

Figure S9: Accumulation of grant money by applicants

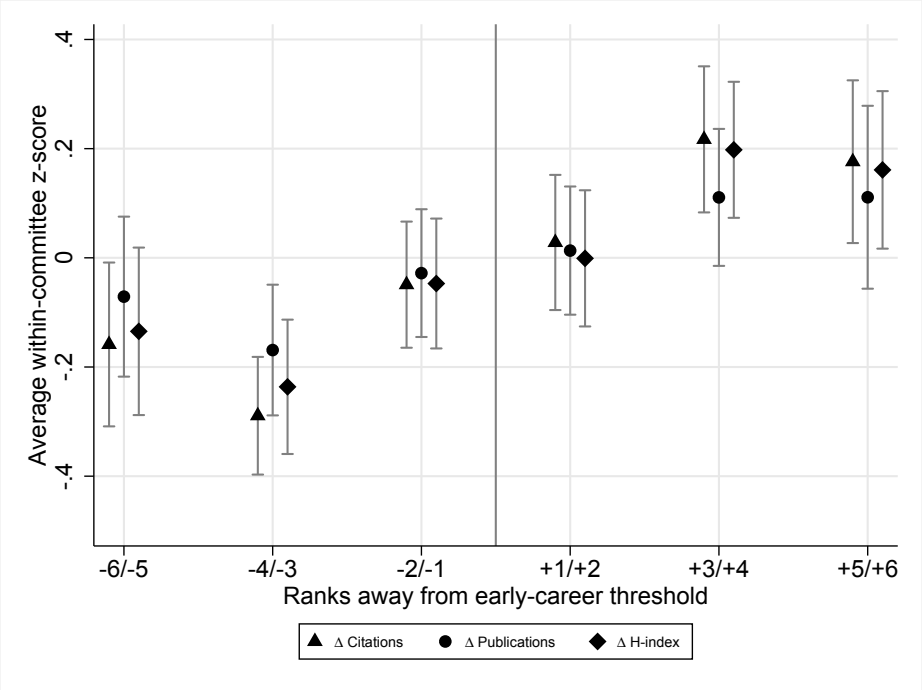


Average growth in research output

In Figure 3 in the main text we find that the lowest ranked winners did not see a larger increase in their output *after* winning the Veni-grant than the highest ranked non-winners. Here we analyze the difference (Δ) between pre-Veni and post-Veni publications, cita-

tions, and H-index. For each applicant we have calculated these differences and then standardized them within each committee to correct for the different publication cultures across disciplines. Figure S10 shows the average standardized differences for the rank groups. The lowest-ranked applicants in our sample score lowest, indicating that their growth in publications, citations, and H-index was lowest. Again, the figure makes clear that there is no significant difference around the cutoff. Moreover, if a change in productivity would explain the larger Vidi probability for +1/+2 versus -1/-2, we would observe a jump around the threshold. Figure S10 makes clear that there is no jump, supporting our conclusions as presented in Figure 4 and Table S8.

Figure S10: Difference between pre- and post Veni output

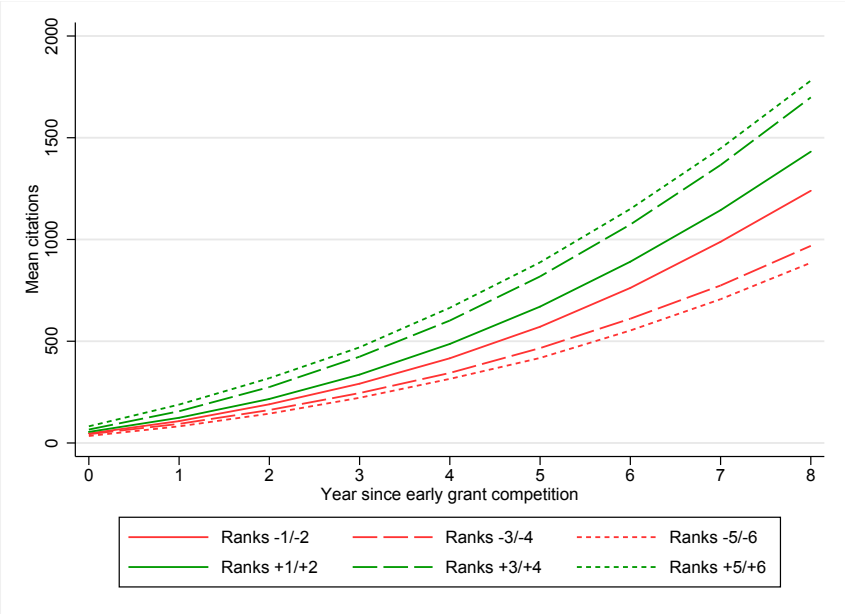


Accumulation of citations

In Figure S11 we investigate if the winners and non-winners diverge in their citations after more than 5 year (Figure 2 in the main text). We plot the mean citations[7] for the 6 rank groups. For these analyses we have weighted all citations to the citations of medical scientists (ZonMW, see Table S2), in order to account for different citation cultures across disciplines. We find that there is a diverging pattern in the mean citations for the different ranks, with the highest scoring applicants (+5/+6) outperforming their neighboring rank (+3/+4) and so on. However, Figure S11 also clearly indicates that there is no jump around the funding cutoff. Higher scoring applicants did perform better when applying for the Veni grant (year 0) and the inequality in citation counts among applicants slowly increases over time, but there is no added effect of winning the grant.

[7] The same results are found when using the median of citations or the mean of the natural logarithm *ln* of citations.

Figure S11: Cumulative citations for different ranks

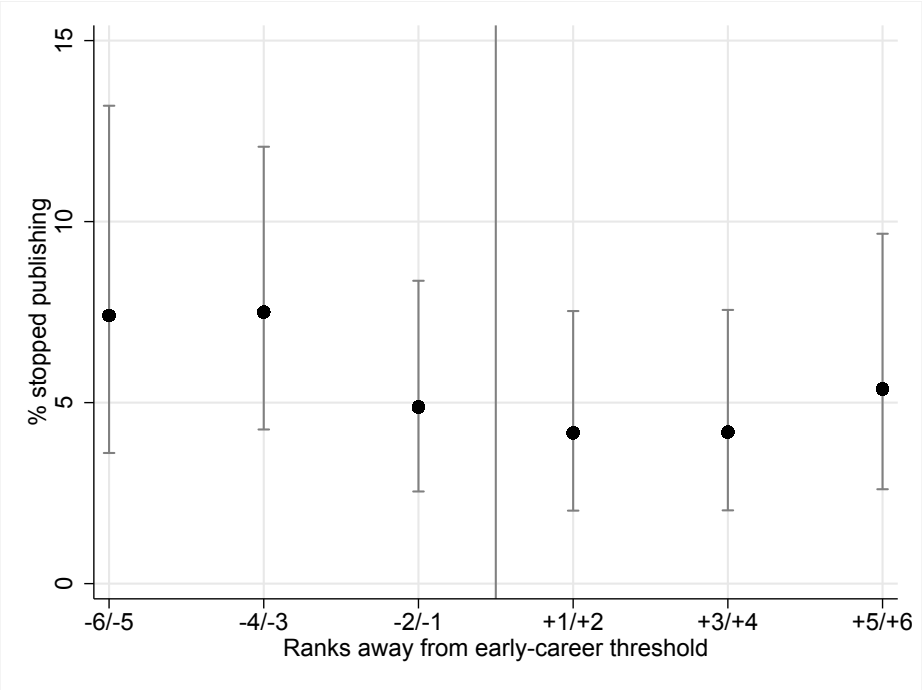


Leaving academia or migrating

In Figure 4 in the main text we find that highest-ranked non-winners are far less likely to apply than lowest-ranked winners. An obvious explanation for this might be that (1) non-winners leave academia or (2) that non-winners migrate after losing the grant[10]. In Figure S12 we test the first explanation by looking at the % of applicants that did not publish any article from year 3-5 after the Veni, but they did publish in the pre-Veni period. This is not a perfect measure, but if non-winners would be more likely to leave academia, we would expect to see a discontinuity. We do not find a discontinuity, suggesting that non-applying non-winners are as likely to work in academia as Veni-winners, but just do not apply to the Vidi. Figure S13 plots the % of applicants at each Veni-rank that is not employed in the Netherlands in early 2017. The information was obtained from the Scopus-profiles of the applicants. Scopus-information on employers is not historical,[8] but again we would expect to see a large difference around the cutoff if migration would be the main reason why non-winners would not apply. We find a small (non-significant) difference between -1/-2 and +1/+2, and the pattern does not indicate a large discontinuity around the cutoff that can explain why the highest ranked non-winners are so much less likely to apply for a Vidi than the lowest-ranked winners.

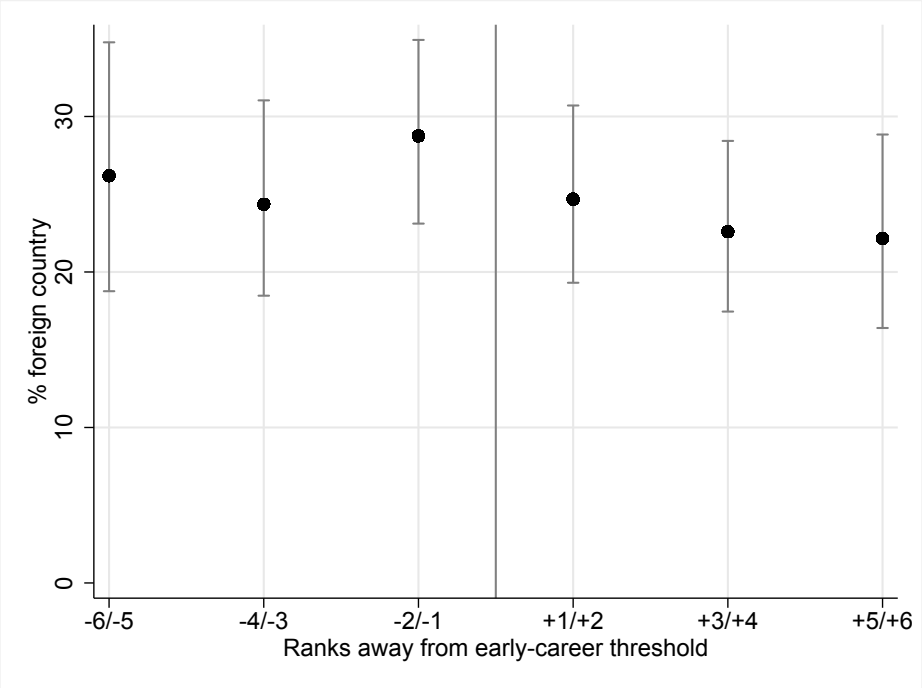
[8]We have obtained employer information from Scopus in early 2017. If authors have multiple affiliations and one of them is the Netherlands, they are defined as working in the Netherlands.

Figure S12: Stopping with publishing after the Veni-application



Note: Applicants are categorized as stopped publishing if they have not published anything 3-5 years after the Veni, but they did publish something before the Veni application.

Figure S13: Working at a foreign university (2017)



Note: Applicants are categorized as not working in the Netherlands if their 2017 Scopus data indicates a non-Dutch employer.

Shifting research direction

An alternative explanation for the main effect we observe is based on the idea that the outcome of the grant competition affects the type of research that winners and non-winners conduct in the following years. Research has indicated that innovative research is less likely to get funded than more established research[11]. If winners receive positive feedback about their research they may be more likely to continue to work on the research they initially proposed to conduct and won them the Veni-grant. Non-winners, however, receive less positive feedback, which may cause them to initiate new types of research. If consistency instead of innovation in research is valued in future grant competitions, this could explain the Veni winner's increased probability of winning a Vidi grant.

To test this alternative mechanism, we evaluate the Vidi project titles of Veni applicants applying for a Vidi grant and we evaluate abstracts of published research before and after the Veni competition. In particular, we examine whether (1) there is greater similarity between project titles of Veni grant applications and project titles of Vidi-grant applications among Veni grant winners than among Veni grant non-winners and whether (2) the abstracts of published work prior to the Veni grant application are more similar to abstracts of published work after the Veni grant application if the focal individual won a Veni grant.

We pre-processed the raw text of each project title and abstract (referred to as texts hereafter) by removing numbers, punctuation, and common English stop words. We then created unigram document-term matrices for each text and for the corpus of combined texts. These matrices record the raw frequencies with which each unigram is present in each text.

Next, we transformed the raw frequencies using the term frequency-inverse document frequency (TF-IDF) statistic. This measure aims to capture the importance of a term to a document in a corpus of texts. It is often used in text mining and information retrieval to weight terms by their frequency within a document, while controlling for their frequency in the entire corpus. An important consequence of this process is the downplaying of common words; however, TF-IDF also provides a more nuanced ranking of terms at both the top and the bottom end of term frequency. The TF-IDF statistic is composed of two parts: term frequency and inverse document frequency. Term frequency (TF) is a straightforward measure of the frequency with which a term appears in a text. Inverse document frequency (IDF) is a measure of the uniqueness of a term in the corpus. It serves to scale up the weights of rare terms and scale down the weights of common terms. Mathematically, it is a logarithmically-scaled measure of the total number of documents divided by the number of documents containing the term — i.e., the inverse relative document frequency. The full TF-IDF statistic itself is simply the product of these two terms.

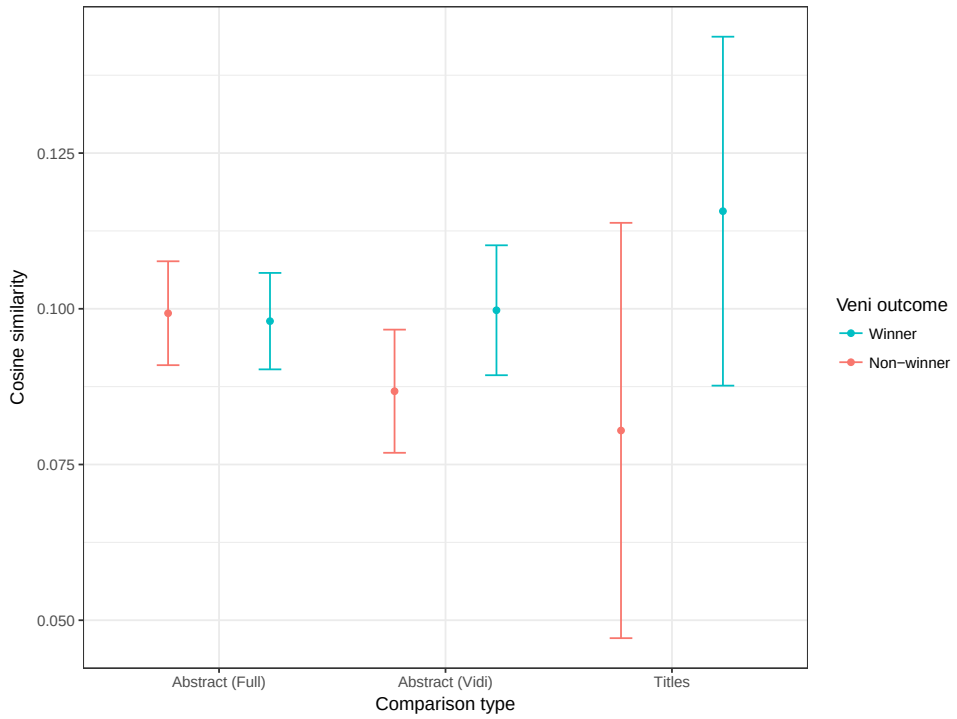
Using these term-text frequencies and TF-IDF statistics, we analyzed pre- and post-

application word usage within Veni grant applicants and compared winners to non-winners. To do this, we employed the cosine similarity metric, $cos(\theta)$, which compares the contents of two vectors, A and B.

$$cos(\theta) = \frac{A \cdot B}{||A||||B||}$$

In this case, because we are comparing two texts in a text corpus, vectors A and B are equivalent to the term frequency vectors of the texts. More specifically, these vectors correspond to the TF-IDF scores contained in the rows of our term-text matrices. In our analyses that focus on project titles we are comparing Veni grant project titles to Vidi grant project titles, whereas our abstract analyses focus on the similarity in abstracts of papers published in the $t-3$ to $t-1$ period and the $t+1$ to $t+3$ period. Here, t refers to the year in which an applicant applied to the Veni grant competition.

Figure S14: Pre and post Veni text comparisons



Results of our analyses are shown in Figure S14. The y-axis represents the cosine similarity, while the x-axis groups three different comparisons. The first comparison ("Abstract (Full)") samples all Veni applicants who submitted an application and were ranked -3 to +3. For each of these applicants, we computed the cosine similarities of the abstracts of published papers in the years preceding the Veni grant application and in the years following the Veni grant application. We then averaged these cosine similarities and compared the averaged between Veni grant winners and Veni grant non-winners. There are no significant differences.

In our next comparison ("Abstract (Vidi)") we limit the sample to Veni applicants who applied for the Vidi grant. Although the graph shows a difference in the point estimates, the difference is not significant. Finally, we compared the project titles of Veni applicants who applied for a Vidi grant. We find that the similarity between Veni and Vidi project titles is slightly higher for winners than non-winners, but like the other comparisons, we find that the difference is not significant. In sum, these findings suggest that the increased likelihood of winning a Vidi grant after winning a Veni grant is unlikely to be the result of non-winners adjusting their research focus.

Shifting to interdisciplinary research

A recent study found that interdisciplinary researchers are less successful in funding competitions[12]. This raises the possibility that the lower funding chances of Veni non-winners in the Vidi competition relative to Veni winners are caused by a shift to interdisciplinary research. In the analyses we investigate if the observed effect in Figure 1 is mediated by the interdisciplinarity of applicants.

We operationalize the level of interdisciplinarity of applicants in terms of the Scopus subject areas of the journals they published during the five years between the Veni and Vidi (t,$t+5$). Scopus defines 27 sub disciplines[9]. We employ two groupings of interdisciplinarity (1) using all 27 sub disciplines, and (2) grouping the 27 sub disciplines in 6 discipline groups. The second grouping takes into account that some combinations of disciplines (e.g., Social Sciences and Psychology) are much more likely than others (e.g., Social Sciences and Physics)[12]. The 6 discipline groups are: Medical Sciences, Biology and Chemistry, Business, STEM fields, Social Sciences and Humanities, and Multidisciplinary.[10] We then calculated the diversity of an author's publications using three indices: (a) Theil's Entropy Index (E), (b) Dissimilarity index (D), and (c) Simpson's interaction index (I)[13]. All articles are given a weight of 1, that was proportionally divided over multiple subject areas if a journal had multiple areas.[11]

We show results only for Theil's Entropy Index E because the reported findings are exactly the same when using the alternative measures (D or I). Theil's Entropy Index is a measure of diversity: the more evenly spread an applicant's articles are over the six

[9]Dentistry (DENT), Health (HEAL), Immunology and Microbiology (IMMU), Medicine (MEDI), Veterinary (VETE), Pharmacology (PHAR), Neuroscience (NEUR), Nursing (NURS), Chemical Engineering (CENG), Chemistry (CHEM), Biochemistry (BIOC), Agricultural and Biological Sciences (AGRI), Business (BUSI), Decision Sciences (DECI), Economics (ECON), Environmental Science (ENVI), Earth and Planetary Science (EART), Energy (ENER), Physics and Astronomy (PHYS), Computer Science (COMP), Mathematics (MATH), Materials Science (MATE), Engineering (ENGI), Arts and Humanities (ARTS), Psychology (PSYC), Social Sciences (SOCI), and Multidisciplinary (MULT)).

[10]The recoding uses the following crosswalk: Medical Sciences (DENT, HEAL, IMMU, MEDI, VETE, PHAR, NEUR, NURS), Biology and Chemistry (CENG, CHEM, BIOC, AGRI), Business (BUSI, DECI, ECON), STEM fields (ENVI, EART, ENER, PHYS, COMP, MATH, MATE, ENGI), Social Sciences and Humanities (ARTS, PSYC, SOCI), and Multidisciplinary (MULT).

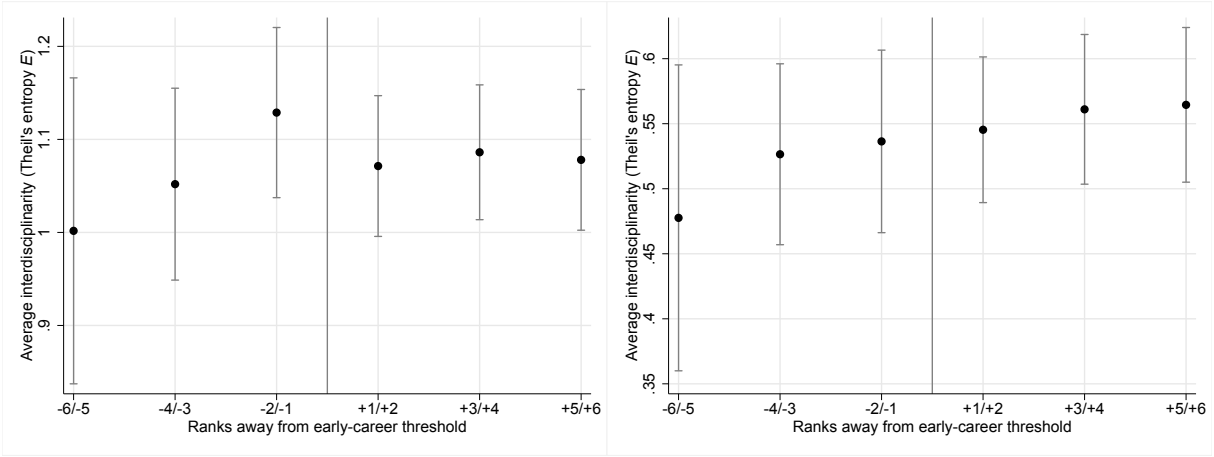
[11]If the subject areas of a journal in which an article was published listed by PSYC and ECON, we gave 0.5 weight to each discipline group.

groups, the higher his or her interdisciplinarity score is. The formal notation is:

$$E = \sum_i p_i ln(\frac{1}{p_i})$$

where i is subject area and p_i the proportion of articles published in that discipline. We calculate E for each applicant separately. Subject area i has a maximum of 27 when using all 27 disciplines and i has a maximum of 6 in the more condensed grouping. E is highest when an applicant's articles are evenly spread across the 27 (or 6) groups and lowest when all articles are published in only one area.

Figure S15: Interdisciplinarity of mid-career garant (Vidi) applicants



(a) Using 27 disciplines

(b) Using 6 discipline groups

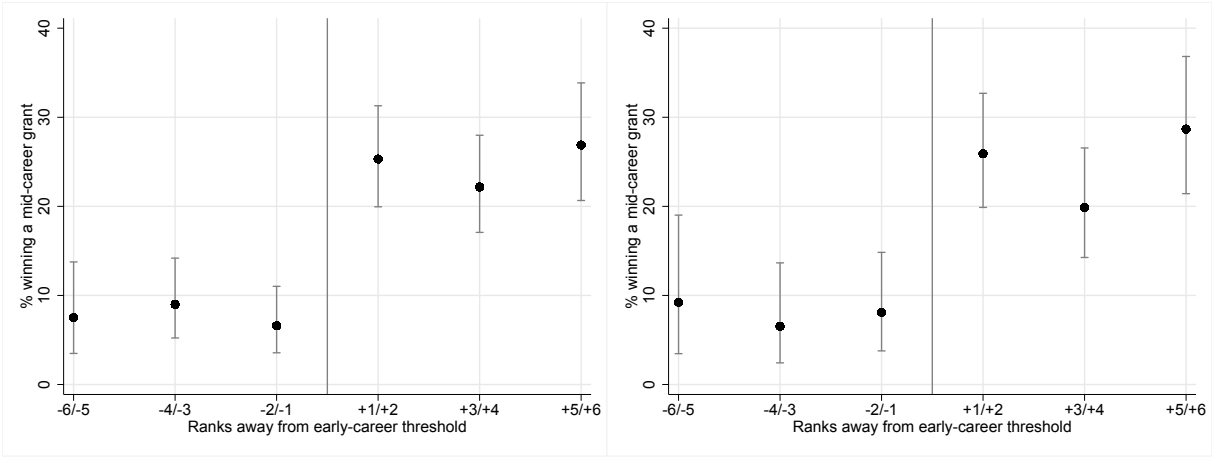
Figure S15 presents the results of the analysis. Both using 27 disciplines (Figure S15a) and using 6 discipline groups (Figure S15b), we find that the average interdisciplinarity of Vidi-applicants does not vary systematically across applicants around the Veni funding cutoff. None of the differences are statistically significant at the 95% level. Had our findings been driven by non-winners being more interdisciplinary at the time of the Vidi application, we would have expected to see a higher interdisciplinary score for the ranks below the threshold.[12] Instead, these results indicate that the Matthew effect in science funding does not operate through a shift by non-winners toward less fundable interdisciplinary proposals.

[12] Additional analyses show that there are no differences in the level of interdisciplinarity of applicants at the time of the Veni-application. This does not correspond with recent findings[12]. These different results might be explained by interdisciplinary applicants not even making it to the interviews. If this would be the case, we would not observe them in our sample around the threshold. Interdisciplinary researchers could be less likely to receive funding[12], but this does not explain why the accumulation of science funding by early career grant winners.

Re-applications

Our final check of an alternative research design choice concerns the sample of applicants. As mentioned in the main text, applicants for the Veni-grant are allowed to apply up to three times. Some of the applicants that were awarded the grant in their second or third try will therefore enter the sample both as non-winner and as winner. In the analyses in the main text, we included those individuals as many times as they applied for the Veni-grant. In Figure S16, Panel A we remove the cases of non-winners who later won the Veni-grant. In Panel B, we remove all cases of re-applicants. The plots clearly show that the discontinuity remains.

Figure S16: Robustness check for re-applicants



(a) Without non-winners who later won

(b) Without all re-applicants

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